

<u>SOFTWARE TESTING</u> [As per Choice Based Credit System(CBCS) &OBE Scheme] SEMESTER–VI			
Course Code:	P22/21IS601	Credits:	03
Teaching Hours/Week(L:T:P):	3:0:0	CIE Marks:	50
Total Theory Teaching Hours:	40	SEE Marks:	50
Course Learning Objectives: This course will enable the students to: • Understand the basics of Software Testing • Recognize various types of Software Testing Techniques • Analyze various levels of software testing • Explore software test automation process			
UNIT– I			8 Hours
Basics of software testing - Basic Definitions, Test Cases, Insights from a Venn Diagram, Identifying Test Cases, Fault Taxonomies, Levels of Testing; Examples: Generalized Pseudo code, The Triangle Problem, The Next Date Function, The Commission Problem			
UNIT– II			8 Hours
Boundary value testing, Equivalence class testing, Decision table based testing -Normal, Robust and Worst-case Boundary value testing, special value testing, Examples, Random testing, Traditional and Improved Equivalence class testing, Equivalence class test cases for triangle problem, NextDate function and commission problem, Decision tables, Decision Table Techniques.			
UNIT– III			8 Hours
Path Testing, Data flow testing, Life Cycle–Based Testing -Program Graphs, DD Paths, Test coverage metrics, Basis path testing, Define/Use Testing, Slice-Based Testing, Program Slicing Tools, Traditional Waterfall Testing, Testing in Iterative Life Cycles			
UNIT– IV			8 Hours
Integration Testing, System testing -Decomposition-Based Integration, Path-Based Integration Example: integration Next Date; Threads, Basis Concepts for Requirements Specification, Model-Based Threads, Use Case–Based Threads, Coverage Metrics for System Testing, Non-functional System Testing			
UNIT– V			8 Hours
Test Management and Automation -Preparing a Test Plan, Scope Management, Deciding Test Approach, Setting up Criteria for testing; Test Automation and terms used, Skills needed for Automation, Scope of Automation, Process Model for Automation, Selecting a Test tool Introduction to Selenium -Overview and working with Selenium IDE, Selenium WebDriver			
COs	Course Outcomes with <i>Action verbs</i> for the Course topics		
CO1	Demonstrate the fundamentals of software testing using real world examples		

CO2	Identify and apply relevant testing techniques suitable for a real world scenario
CO3	Apply the different levels in testing
CO4	Implement Test Automation process and experiment with testing tools
Text Book(s): 1 PaulC.Jorgensen,“SoftwareTesting,ACraftsman’sApproach”,AuerbachPublications, 3rd Edition, First Indian Reprint, 2014, ISBN-13:9781466560680 2 SrinivasanDesikanGopalaswamy,“SoftwareTestingPrinciplesandPractices”,Pearson Education, 5thEdition, 2008 , ISBN: 9788177581218	
Reference Book(s): 1. UnmeshGunecha,“LearningSeleniumTestingToolswithPython”,PACKTPublishing, 2014,ISBN: ISBN 9781783983506	
Web and Video link(s): 1. https://artoftesting.com/selenium-tutorial 2. https://katalon.com/resources-center/blog/software-testing 3. https://www.tutorialspoint.com/software_testing/index.htm	
E-Books/Resources: 1. https://books.google.mn/books?id=Yt2yRW6du9wC&printsec=copyright&source=gbs_pu_b_info_r#v=onepage&q&f=false 2. https://www.tutorialspoint.com/selenium/selenium_pdf_version.htm	

- 1 PaulC.Jorgensen,“SoftwareTesting,ACraftsman’sApproach”,AuerbachPublications, 3rd
Edition, First Indian Reprint, 2014, ISBN-13:9781466560680
- 2 SrinivasanDesikanGopalswamy,“SoftwareTestingPrinciplesandPractices”,Pearson
Education, 5thEdition, 2008 , ISBN: 9788177581218

1. UnmeshGunecha,“LearningSeleniumTestingToolswithPython”,PACKTPublishing, 2014,ISBN: ISBN 9781783983506

1. <https://artoftesting.com/selenium-tutorial>
2. <https://katalon.com/resources-center/blog/software-testing>
3. https://www.tutorialspoint.com/software_testing/index.htm

1. https://books.google.mn/books?id=Yt2yRW6du9wC&printsec=copyright&source=gbs_pu_b_info_r#v=onepage&q&f=false
2. https://www.tutorialspoint.com/selenium/selenium_pdf_version.htm

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